Software Engineer Assessment: ADM Document Automation

Build the pipeline that turns client data into a publication-ready strategy document — automatically.

ROLE

Software Engineer

Overview

We produce **Account Development Master (ADM) documents** — detailed, data-rich strategy plans — for large enterprise clients. Each document covers 12 sections: executive summary, portfolio analysis, app inventory, competitive benchmarking, AI transformation strategy, modernization roadmap, cloud & data strategy, financials, execution roadmap, delivery center architecture, benchmarking summary, and partnership overview.

Today this is done entirely by hand. A consultant gathers client information, writes a detailed prompt, runs an AI model across 12 consecutive batches, and assembles the final HTML output manually. It takes hours per client. **Your job is to automate this end-to-end.**

We have attached two things for reference: a **sample ADM document** (Cisco Systems) showing exactly what the final output should look like, and the **master prompt** we currently use to generate it manually. Both are your blueprint.

What You Need to Build

A system that accepts client business information as input and produces a complete, publication-ready ADM document as a self-contained HTML file — automatically, matching the structure, depth, and visual quality of the attached sample.

1 · Client Input Model

Define a structured data model for all client information needed to generate an ADM document. This includes the company profile, annual ADM spend, application inventory (each app's name, owning business unit, age, tech stack, and annual run cost), top competitors with known capability gaps, preferred delivery center locations, and transformation targets. This model is the single source of truth that drives everything else.

2 · Financial Calculation Layer

Build a dedicated calculation module that derives all key financial figures directly from the client input — total 5-year contract value, cumulative business value, target ROI, workforce savings from offshore rate arbitrage, year-by-year investment phasing, and the legacy cost reduction trajectory across the programme. These numbers must be computed by your code. The AI model must never be asked to generate or estimate financial figures — it receives them as inputs and weaves them into narrative.

3 · AI Generation Pipeline

Using the attached master prompt as your reference, build a pipeline that calls an AI model to generate each of the 12 document sections in sequence. Each section call must inject the pre-computed financials and relevant client data so the output is specific, analytical, and numerically accurate — not generic. The pipeline must handle failures gracefully: if a section call fails, it should retry automatically without restarting the entire run from scratch.

4 · HTML Document Renderer

Assemble the AI-generated section content into a final HTML document that mirrors the attached sample. The output must include all the same components: the headline KPI grid on the cover, application inventory table, competitive benchmarking comparison cards, the six-column modernization matrix (Retire / Retain / Rehost / Replatform / Refactor / Rearchitect), a 5-year financial chart, the year-by-year execution roadmap, and the delivery center layout. The document must be fully self-contained — a single HTML file that works when opened in any browser — with a working sidebar that navigates between all 12 sections.

5 · End-to-End Test with a New Client

Run your complete pipeline against a fictional client in a different industry from Cisco — financial services, healthcare, retail, or manufacturing all work. The generated ADM document for this client is the primary thing we will review. It should be substantive and specific to that client, with no placeholder text, no generic filler, and financial figures that are consistent and accurate throughout. Share the generated HTML file and the client input data you used to produce it.

The core principle: The AI generates language. Your code generates numbers. Every financial figure in the document — ROI, TCV, investment by year, cost savings — must trace back to a calculation in your codebase, not to something the model decided on its own.

What to Share With Us

When you're done, share a private GitHub repository containing your full codebase, the generated ADM HTML file for your fictional client, and the client input data you used. Include a short README explaining how to set it up and run it for a different client from scratch.